

HLS synthesized latency:

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== Performance Estimates
=====
+ Timing:
  * Summary:
  +-----+-----+-----+-----+
  | Clock | Target | Estimated| Uncertainty|
  +-----+-----+-----+-----+
  | ap_clk | 10.00 ns| 7.300 ns| 2.70 ns|
  +-----+-----+-----+-----+

+ Latency:
  * Summary:
  +-----+-----+-----+-----+-----+-----+
  | Latency (cycles) | Latency (absolute) | Interval | Pipeline |
  | min | max | min | max | min | max | Type |
  +-----+-----+-----+-----+-----+-----+
  | 95273 | 95518 | 0.953 ms | 0.955 ms | 81666 | 95490 | dataflow|
  +-----+-----+-----+-----+-----+-----+

```

Co-simulation latency and clock period:

RTL	Status	Latency(Clock Cycles)			Interval(Clock Cycles)			Total Execution Time (Clock Cycles)
		min	avg	max	min	avg	max	
VHDL	NA	NA	NA	NA	NA	NA	NA	NA
Verilog	Pass	81966	81966	81966	NA	NA	NA	81966

Post-implementation resource utilization:

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#=== Post-Implementation Resource usage ===
SLICE:          0
LUT:           5113
FF:            5435
DSP:           58
BRAM:          13
URAM:          0
LATCH:         0
SRL:           145
CLB:           1302

#=== Final timing ===
CP required:          10.000
CP achieved post-synthesis: 4.840
CP achieved post-implementation: 6.547
Timing met

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Final speedup ratio calculation:

81966 cycles * 6.547 post-imp. clock period = 0.5366 ms

51.962 / 0.5366 = 96.83 speedup